TR30 example 7: high-precision appendix

KISSsoft 2026 / ISO 6336:2019. M33 remains UNACCEPTED. Pair with the complete vendor PDF/RTF and native Z12. Values below are COM readbacks after saving, releasing, reopening and recalculating. CSV retains 17 significant digits.

Quantity	Gear 1	Gear 2
YF	0.857076971287	0.893702035141
YS	2.2973660385	2.35760033491
sigmaF0 (MPa)	48.0185351217	51.3832790602
sigmaF (MPa)	150.57774022	161.128989595
sigmaFP (MPa)	420.374917958	403.26934791
SF	2.79174675714	2.50277339245
sigmaH (MPa)	374.780325458	374.780325458
sigmaHP (MPa)	488.39894355	499.004417741
SH	1.30316057267	1.33145841402

Hardness conversion settings

Material = Own Input; through-hardened alloy steel; ML (mat.isoclass=1, verified in material UI). Entered surface hardness unit is HBW (mat.haerteDim=0). Gear 1 is 342 HBW; gear 2 is 300 HBW. No HV value was substituted for the table input.

Quantity	Gear 1	Gear 2
mat.haerte	342	300
HardnessHB	342	300
HardnessHV	359.810625986	315.623356128
mat.limh	660.43135192	602.413466597
mat.limf	256.199894792	237.508679642

The gear-module readbacks are 359.8106259863 HV and 315.6233561284 HV. These two points equal HB/0.9505 numerically; this observation does not establish the vendor algorithm. The help page describes ISO 18265:2014 Table A.1 for the material-window conversion, whose adjacent-point interpolation gives 315.5555555556 HV at 300 HB. These are recorded as different paths; no claim of equivalence is made.

Material limits were explicitly entered from the ISO 6336-5 V-alloy ML formulas using the module HV: sigmaHlim=1.313*HV+188; sigmaFlim=0.423*HV+104. KISSsoft independently calculates geometry, factors and stresses; this is not a fully independent material-limit validation.

Input preservation and comparison limits

All original table rows are listed in input-audit.csv, including dashes and method selections. Measured span is mapped to tooth-thickness allowance by As=(Wtable-Wnominal)/cos(alpha_n). Both allowance limits are set equal; the four spans read back exactly. The parenthesized x2 values are dependent geometry values and agree at the printed precision. For example 7, KISSsoft width includes the 50.8 mm gap: 215.9 mm overall, 165.1 mm active total, 82.55 mm per helix.

An unresolved representation assumption remains: TR30 gives tip relief Ca=13 um but no relief length or shape. KISSsoft overwrites the derived Ca field unless a native modification is present. These files use symmetric linear relief, C=13 um, LCa=0.5*mn, with profile-deflection compensation still disabled. LCa=1*mn is checked separately for sensitivity of every mapped comparison output. Neither length is attributed to TR30, and strict input closure remains false.

Oil nu40 is entered exactly (25 / 220 mm2/s). Other oil thermal, scuffing and micropitting settings inherited from the vendor seed are ancillary defaults absent from TR30; those extra report sections are not TR30 validation evidence. Surface-condition assumptions and any unavailable software quantities remain explicit; M33 is not accepted.

Rows below show remaining differences against printed reference values using the pre-existing v0.8 display-quantum diagnostic. The strict absolute 0.01 AND relative 0.5% columns are separately preserved in comparison.csv. Neither diagnostic grants acceptance.

Quantity	TR30 printed	KISSsoft	Difference
sigma_FP_1	421	420.374918	-0.62508204
sigma_FP_2	404	403.2693479	-0.73065209
sigma_Hlim_1	661	660.4313519	-0.56864808
sigma_Hlim_2	603	602.4134666	-0.5865334
sigma_HP_1	489	488.3989435	-0.60105645

Full line-by-line comparison (including gearmath v0.8 with its original formulas and material conversion) is in comparison.csv / comparison.json. f_sh0 is not exposed in this extraction; no derived convention was silently substituted. Input checks: {'matched': 109, 'not_applicable': 20, 'derived_matches_printed': 1}. Reopen numeric changes: 0.

COM calculation messages were empty, but opening example 7 in the GUI displayed informational warnings about fZCa selection, high circumferential speed/lubrication, scuffing method range and feasibility of span measurement on gear 2. The original table values were retained. Example 8 required an initial second calculation without changing inputs; final and reopened calculations returned true. See run.json for each attempt.